

ZEN-Garden PV supply chain

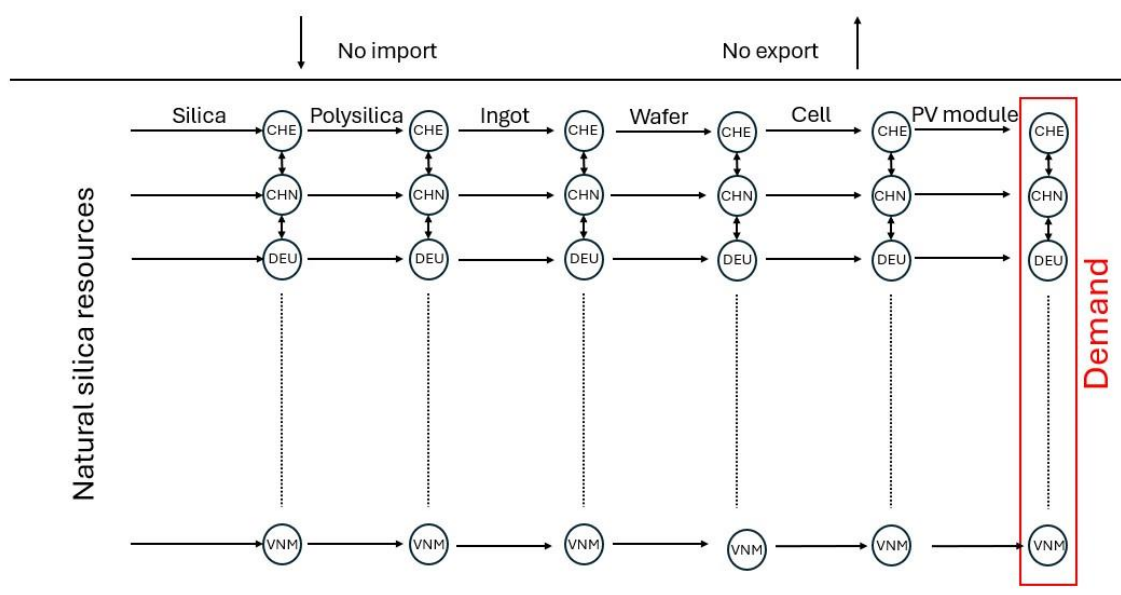
Folder structure:

All results and data are gathered in the MatthiasSPData folder. The results obtained and graphs generated from the specific project can be found in the corresponding folders. Everything related to the model can be found in the ZEN-garden folder. There are all input parameters defined and the model definition.

ZEN-Garden:

ZEN-garden is a model-based approach to model the energy system. This model is changed to a model for PV supply chains. The biggest change is the change to a physical product and the characteristic coming with the change.

The Model for supply chains have been changed from a model where each node is representing a unit with Input/Output and conversion technologies to a model where each production step represents a layer. In each layer trade can be done. In general, it is a different thinking about the model, but it can be displayed in both ways.



Import and export is set to zero as import is considered as incoming material from outside the model. Transport from one node to the next is not considered as import of the buying node but as transport. Silica is modeled as Import and import availability is set to natural resource per country. Demand is the triggering parameter for the supply chain. There is no other demand.

An introduction to the ZEN-garden environment can be found here: https://github.com/ZEN-universe/ZEN-garden/blob/main/documentation/how_to_ZEN_garden.md

It is important to install the ZEN-garden package properly. This can take some time. For the simulation of the model, I used the anaconda prompt option. While working in the ZEN-garden environment ensure to be always in the ZEN-garden environment. You can check this with the name in brackets in front of the cmd line (activate with: anaconda activate zen-garden). For

doing so it is important to have the right folder structure and navigate in the anaconda cmd with `cd` (change directory) to the Data folder. Once your in the right folder you can start the project with: `python -m zen_garden --dataset="Name_of_Folder_with_Model"`

The simulation can take some time.

Input:

The input parameters are a key element of the ZEN-garden model. In the .json file each input parameter is defined by the name the unit and the default value. To change the default value a csv file with the exact name can be created and placed in the same folder. The csv file consists of a header, which can vary from input variable to input variable. The correct header can be found by the already existing files, the example files from the GitHub ZEN-garden introduction examples or asked by a more experienced ZEN-garden developer. After the header the values for each time, node or edge can be defined (as defined in the header).

 attributes	09.05.2024 18:10	JSON-Quelldatei	2 KB
 capacity_existing	18.04.2024 16:18	Microsoft Excel-CSV-...	1 KB
 capacity_limit	10.05.2024 15:17	Microsoft Excel-CSV-...	2 KB
 capex_specific	09.04.2024 15:44	Microsoft Excel-CSV-...	1 KB
 capex_specific_CHE	02.05.2024 11:13	Microsoft Excel-CSV-...	1 KB
 capex_specific_EU	02.05.2024 11:15	Microsoft Excel-CSV-...	1 KB
 capex_specific_EUUSA	02.05.2024 11:15	Microsoft Excel-CSV-...	1 KB
 capex_specific_world	02.05.2024 11:16	Microsoft Excel-CSV-...	1 KB
 carbon_intensity_technology	19.04.2024 09:36	Microsoft Excel-CSV-...	1 KB
 employee_technology	10.05.2024 16:17	Microsoft Excel-CSV-...	1 KB
 opex_specific_fixed	09.04.2024 16:34	Microsoft Excel-CSV-...	2 KB
 opex_specific_fixed_CHE	02.05.2024 11:48	Microsoft Excel-CSV-...	2 KB
 opex_specific_fixed_EU	02.05.2024 11:51	Microsoft Excel-CSV-...	2 KB
 opex_specific_fixed_EUUSA	02.05.2024 11:53	Microsoft Excel-CSV-...	2 KB
 opex_specific_fixed_world	02.05.2024 11:54	Microsoft Excel-CSV-...	2 KB

This is a example of input files for a conversion technology with different scenarios.

At the very end of the provided link to the ZEN-garden environment a list with all input and output parameters and more is provided.

Output:

The output can be generated with the result file and the parameters defined in ZEN-garden. The output is given as csv fille. The results file in the Data folder is generating the output. In the actual project the important output parameters are generated and then saved in the local file structure.

Objective:

Objectives are always defined as minimization problems. They are defined in the *ZEN-garden\zen_garden\model\objects* Energy_system file. They are always optimitzing over the whole design period. The security and employee objectives are created in the last project based on the sustainability objective. Also new parameters were defined for the new objectives those must be added to all .json files.

Initially by defining the parameters the unit is handed over to the parameter. This must be equal to the unit of the .json file.

In the project four objectives are defined.

- Total cost -> minimizing the total global cost over the modeled period
- Sustainability -> minimizing the total global emissions
- Employee -> maximizing the number of employees (reciprocal to minimize)
- Security -> minimizing the number of trades

Improvements/How to go on:

To improve the model following options are available. Add number of employees and CAPEX costs to transport. Add taxes to the transport. Avoid circumventions in the trade restriction scenarios. With import costs the mining cost for each country can be modeled. Input data can be refined. Distance between nodes optimized and the costs calculated more properly. More nodes can be imposed.

The trade objective and the employee objective can be improved since it is minimizing one value per conversion technology which is the same for a lot of countries. The results are to straight forward. This means employees for trade and better input data can be used.

To improve the security objective the transport routes can be weighted by a risk. For example, the trade between USA and China has penalty 5 because of trade war. Trade between Switzerland and Germany can be modeled as one because high stability.

Other:

Don't forget to adjust names and file structures.